

COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No: 01

Miscellaneous Topics



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Recap of Previous Lecture



Question Discussion



Topics to be Covered

Number System



[Number System]



Base of a number system : $(r) \longrightarrow [0-1- \dots -(r-1)]$ digits

decimal no. system $\longrightarrow$ base $= r = 10 \longrightarrow (0-1- \dots 9)$ digits

$$\begin{array}{ccc} 10^2 & 10^1 & 10^0 \\ (286)_{10} \end{array}$$

$\longrightarrow$ base $= r = 7 \longrightarrow (0-6)$ digit

$$\begin{array}{ccc} (286)_7 & \times \\ 7^2 & 7^1 & 7^0 \\ (326)_7 \end{array}$$

$$\begin{array}{ccc} x^2 & x^1 & x^0 \\ (286)_x \end{array} \longrightarrow \text{for no. to be valid} \longrightarrow \begin{array}{l} x \geq 9 \\ x > 8 \end{array}$$

$$= (2x^2 + 8x + 6)_{10}$$

$$\begin{matrix} 7^1 & 7^0 \\ (2 & 4) \end{matrix}_7 = (14 + 4)_{10} = (18)_{10}$$

$$(18)_{10} = \begin{matrix} 7^1 & 7^0 \\ (2 & 4) \end{matrix}_7$$

$$(77)_{10} = \begin{matrix} 7^2 & 7^1 & 7^0 \\ (1 & 4 & 0) \end{matrix}_7$$

$$(111)_{10} = \begin{matrix} 7^2 & 7^1 & 7^0 \\ (2 & 1 & 6) \end{matrix}_7$$

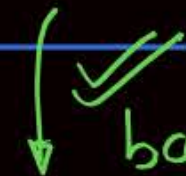
$$(73)_{10} = \begin{matrix} 8^2 & 8^1 & 8^0 \\ (1 & 1 & 1) \end{matrix}_8 = (8^2 \times 1 + 8 \times 1 + 8^0 \times 1)_{10}$$

$$(n \text{ digit})_r \rightarrow$$

$$[(r-1), (r-1) \dots n \text{ times}]_r = (r^n - 1)_{10}$$

$$(1 \ 1 \ 1 \dots n \text{ times})_2 = (2^n - 1)_{10}$$

$$r\text{-base system} \rightarrow n\text{-digits} [0 - (r^n - 1)_{10}]$$



$$\text{base} \rightarrow 7 \rightarrow \text{max no. of 2 digit} = (66)_7$$

$$= (48)_{10} = (7^2 - 1)_{10}$$

$$\text{base} \rightarrow 8 \rightarrow \text{max no. of 2 digit}$$

$$= (77)_8 = (63)_{10} = (8^2 - 1)_{10}$$

- Imp number system and their conversions :

decimal no. system	base = 10	→ (0-9)
binary no. system	base = 2	→ (0-1)
Octal no. system	base = 8	→ (0-7)
Hexadecimal no. system	base = 16	→ (0-15)

(A 10)₁₆

10	→	A
11	→	B
12	→	C
13	→	D
14	→	E
15	→	F

$$(47)_{10} = (101111)_2$$

• Octal no. system $\rightarrow$ base = 8 $\rightarrow$ digits (0-7)

$$(273)_8 \checkmark$$

$$(389)_8 \times$$

$$\bullet \begin{matrix} 8^2 & 8^1 & 8^0 \\ (123)_8 \end{matrix} = (8^2 \times 1 + 8^1 \times 2 + 8^0 \times 3)_{10} = (83)_{10}$$

$$\bullet \begin{matrix} 8^2 & 8^1 & 8^0 \\ (154)_{10} \end{matrix} = (232)_8$$

0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

$$\begin{matrix} (010101)_2 \\ \hline \end{matrix} = (25)_8$$

$$= (21)_{10} \quad = (16+5)_{10}$$

$$(273)_8 = (010\ 111\ 011)_2$$

$$\cdot (\underbrace{001011}_{\text{13}} \cdot \underbrace{101010}_{\text{52}})_2 = (13 \cdot 52)_8$$

$$\cdot (\underbrace{001011101}_{\text{135}} \cdot \underbrace{010110110}_{\text{266}})_2 = (135 \cdot 266)$$

$$\cdot (\underbrace{101110111}_{\text{567}} \cdot \underbrace{011111100}_{\text{374}})_2 = (567 \cdot 374)_8$$



- Hexadecimal no. System:

base = 16 $\rightarrow$ digits (0-15)

$$\begin{matrix} 16^2 & 16^1 & 16^0 \\ (2AB)_{16} = (256 \times 2 + 16 \times 10 + 16^0 \times 11)_{10} \\ = (683)_{10} \end{matrix}$$

10 - A

11 $\rightarrow$ B

12 - C

13 - D

14 - E

15 - F

$$(275)_{10} = \begin{matrix} 16^2 & 16^1 & 16^0 \\ (113)_{16} \end{matrix}$$

$$(207)_{10} = \begin{matrix} 16^1 & 16^0 \\ (CF)_{16} \end{matrix}$$

$$(\underbrace{1011}_{10} \underbrace{0110}_{6})_2 = (B6)_{16}$$

A	9
A	A
A	B
A	C
A	D
A	E

$$AF = \begin{matrix} 1010 & 1111 \\ \hline B & 0 \end{matrix}_2$$

$$\bullet \left(\underbrace{01011011}_{\text{5B}} \cdot \underbrace{10111101}_{\text{BD}} \underbrace{10}_{\text{8}} \right)_2 = (5B.BD8)_{16}$$

$$\bullet \left(\underbrace{11101110}_{\text{1D}} \underbrace{11010}_{\text{DA}} \cdot \underbrace{0111}_{\text{7}} \underbrace{1010}_{\text{AC}} \underbrace{110}_{\text{C}} \right)_2 = (1DDA.7AC)_{16}$$

$$\bullet (2AB.CF)_{16} = (001010101011 \cdot 11001111)_2$$

$$\bullet (26D.64B)_{16} = (001001101101 \cdot 011001001011)_2$$

- $(746)_8 = (1E6)_{16}$

$$\hookrightarrow (\underbrace{111}_{100} \underbrace{100}_{110})_2 = (1E6)_{16}$$

- $(A14)_{16} = (5024)_8$

$$\hookrightarrow (\underbrace{1010}_{0001} \underbrace{0001}_{0100})_2$$

[Question]



A number $N_1 = (142)_5$ then it will be equal to $(xyz)_6$ then $(x + y + z)$ is $(\underline{7})_{10}$.

$$\begin{matrix} 5^2 & 5^1 & 5^0 \\ (142)_5 \end{matrix} = (25 + 20 + 2)_{10} = (47)_{10} = \begin{matrix} 6^2 & 6^1 & 6^0 \\ (1 & 1 & 5) \end{matrix}_6$$

$$x + y + z = 1 + 1 + 5 = (7)_{10}$$

$$\begin{array}{r} 1 \\ + 1 \\ + 5 \\ \hline (11)6 \end{array}$$

[Question]



A number $N_1 = (247.56)_8$ then its hexadecimal equivalent number is

(a) $A7.B7 = (\underbrace{010100}_{\text{A}} \underbrace{111}_{\text{7}} \cdot \underbrace{101110}_{\text{B}})_{\text{2}}$

☒ (b) $A7.B8 = (A7.B8)_{16}$

☐ (c) $(47.B8)$

☐ (d) $B7.B8$

[Question]

A binary no. is given as $B = 101010.110100$ then its hexadecimal and octal equivalent will be :

$$(5 \ 2 \cdot 6 \ 4)_8$$



$$(2A \cdot D)_{16}$$

(a) ~~$(A2.D)_{16}, (52.64)_8$~~

(b) $(2A.D)_{16}, (52.64)_8$

(c) $(2A.D)_{16}, (25.64)_8$

(d) ~~$(A2.D)_{16}, (25.64)_8$~~

[Question]

Two no. N_1 and N_2 are given as

$$N_1 = (111)_2, = (7)_{16} \quad N_2 = (777)_8$$

Then their hexadecimal equivalent will be :

☒ (a) $(7)_{16}, (1FF)_{16}$

(b) $(7)_{16}, (777)_{16}$

(c) ☒ $(F)_{16}, (777)_{16}$

(d) ☒ $(F)_{16}, (1FF)_{16}$

$$= \left(\underbrace{1111}_4 \underbrace{1111}_4 \right)_2 = (1FF)_{16}$$

[Question]



A relation is given as :

$$\overset{x^2 x^1 x^0}{(124)_x} = \overset{y^1 y^0}{(64)_y} \quad \& \quad \checkmark$$

$$x \geq 5 \quad \checkmark$$

$$y \geq 7$$

$$(x+y)$$

infinite solutions.

Then minimum value of $(x+y)$ 14.

$$(x^2 + 2x + 4)_{10} = (6y + 4)_{10}$$

$$y = \left(\frac{x^2 + 2x}{6} \right) = \frac{x(x+2)}{6}$$

$$x=5 \quad (39)_{10}$$

$$y=7 \quad (46)_{10}$$

$$x=6, \quad y=8 \quad x=18, \quad y=60$$

$$x=6 \quad (52)_{10}$$

$$y=8 \quad (52)_{10}$$

$$x=10 \quad y=20$$

$$x=12 \quad y=28$$

$$x=16 \quad y=40$$



Topic : 2 Min Summary

→ Number System



Thank you

GW
Soldiers !

